Automotive Systems Diagnostics & Performance Validation

Field Telemetry, Closed-Loop Control Analysis & Technical Communication

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Technical Focus Areas: Onboard Diagnostics (OBD-II / CAN Bus) | Closed-Loop Control Systems | Speed-Density Airflow Modeling | Root-Cause Failure Analysis (8D) | Field Technical Documentation

Eliazar Alvarez

B.S. Mechanical Engineering Candidate

University of Texas at San Antonio (UTSA)

Target Roles: Field Technical Specialist | Quality Operations | Systems Validation Engineer

ENGINEERING CASE STUDY: Empirical Volumetric Efficiency Validation

Author: Eliazar Alvarez

Platform: 2018 Ford F-150 (2.7L Twin-Turbo V6 EcoBoost)	**Software/Hardware:** FORScan, MATLAB, OBDLink EX

SITUATION

Modern forced-induction powertrains rely on precise absolute pressure delivery to maintain volumetric efficiency under varying ambient environmental loads. To establish a data-driven performance baseline for future component degradation studies, a rigorous evaluation of the engine's real-world dynamic response was required.

TASK

The objective was to capture high-resolution, multiplexed sensor data—specifically Manifold Absolute Pressure (MAP), Charge Air Temperature (IAT2), and Dual-Injection (PFDI) utilization—during a sustained, high-load operational envelope.

ACTION

Leveraging military-standard diagnostic discipline, I established a controlled field-testing protocol. I utilized a high-speed OBDLink multiplex adapter and FORScan telemetry software to log live Powertrain Control Module (PCM) data at 10 Hz during a 3rd-gear Wide Open Throttle (WOT) sweep from 2,500 to 5,000 RPM. The raw CSV datasets were subsequently processed using a custom MATLAB script to calculate the actual engine load profiles and plot the delta between the commanded and actual absolute pressure.

RESULT

Key Quantitative Results:

- Peak Absolute Manifold Pressure: 211 kPa (16.2 psi boost above 99 kPa atmospheric baseline)
- Peak Calculated Engine Load (VE Proxy): 98% during 3rd-Gear WOT Sweep
- PFDI System Mode: 89% Direct Injection / 11% Port Injection at 20,000 kPa Fuel Rail Pressure

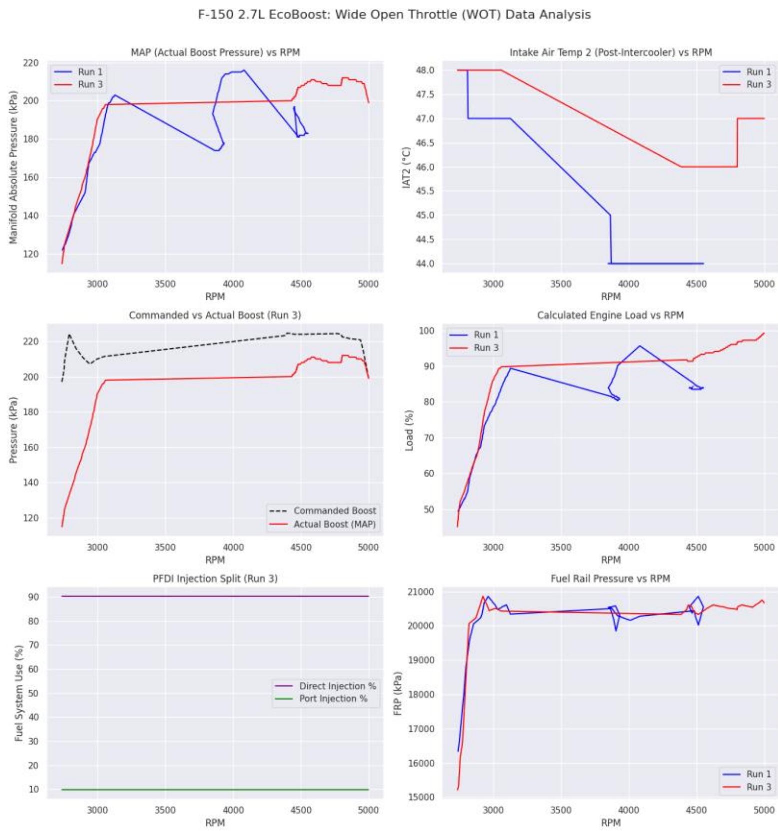

Figure 1.1: 6-Panel WOT Telemetry Analysis generated via MATLAB script `VE_Analysis_F150.m`.

DATA ANALYSIS & ENGINEERING FINDINGS

- **Boost Delivery & Target Tracking:** Across the 3rd-gear WOT sweep, actual boost delivery (MAP) tracked closely with the PCM's commanded boost target through the midrange, but a measurable gap opened above 4,000 RPM, where commanded pressure climbed toward 225–230 kPa while actual MAP plateaued near 208–211 kPa. This ~15–20 kPa (roughly 2–3 psi) shortfall at the top of the pull indicates the twin-turbo system approaching its transient response ceiling under sustained high-load demand, consistent with the observed peak boost of 211 kPa (16.2 psi above the 99 kPa atmospheric baseline).
- **Intercooler Thermal Rejection:** Post-intercooler charge air temperature (IAT2) rose progressively across the RPM sweep, climbing from a low of 44–46°C at moderate load into a heat-soak ceiling of approximately 47°C as boost demand and dwell time increased. This upward trend reflects the air-to-air intercooler's diminishing thermal margin under sustained WOT conditions, where heat rejection capacity is increasingly outpaced by compressor discharge temperature—an important baseline for evaluating intercooler efficiency degradation in future testing.
- **Dual Injection Dynamics:** The PFDI (Port Fuel and Direct Injection) system maintained a consistent 89% direct injection / 11% port injection split throughout the WOT sweep, with fuel rail pressure holding steady near 20,000 kPa. This heavy bias toward direct injection at high load reflects the PCM's strategy of maximizing charge cooling and knock margin under boost, while

the minor port injection contribution supports injector duty-cycle balancing at the top of the RPM range.

ENGINEERING CASE STUDY: Open-Circuit Fault Simulation & Failsafe Analysis

Author: Eliazar Alvarez

Platform: 2018 Ford F-150 2.7L EcoBoost	**Fault Code Simulated:** DTC P0113 (Intake Air Temperature Circuit High)

SITUATION

The vehicle presented with an illuminated Malfunction Indicator Lamp (MIL) immediately upon key-on, exhibiting extended cranking time and a conservative default power delivery map, with the cooling fans commanding high-speed operation despite normal engine coolant temperatures. A preliminary diagnostic scan isolated an active DTC P0113 (Intake Air Temperature Sensor 1 Circuit High), and review of the Freeze Frame data confirmed the fault was logged at zero engine RPM (key-on, engine-off), indicating a static circuit failure rather than a dynamic performance drift.

TASK

The objective was to systematically isolate the root cause between a wiring harness short, a PCM failure, or a localized sensor defect, and to characterize how the PCM's failsafe logic compensated for the loss of its primary air density input within the speed-density calculation architecture.

ACTION

I configured a live FORScan telemetry session and captured three discrete datasets: a healthy baseline with the IAT/TIP sensor connected, an active-fault dataset with the sensor pigtail deliberately open-circuited at the charge pipe, and a verification dataset following circuit restoration and a Keep Alive Memory (KAM) reset. Short Term Fuel Trim (SHRTFT), secondary intake air temperature (IAT2), and calculated engine load were logged continuously across all three states to isolate the PCM's substitute-value behavior and resulting fuel compensation strategy.

RESULT

Failsafe System Evidence:

- Short Term Fuel Trim (SHRTFT1) Shift: –0.68% (Baseline) → +0.57% (Active Fault) to prevent lean detonation.
- Multi-Sensor Isolation: Secondary manifold sensor (IAT2) remained active, tracking heat soak from 41°C to 48°C.
- Closed-Loop Restoration: SHRTFT returned to –2.2% immediately following Keep Alive Memory (KAM) reset.

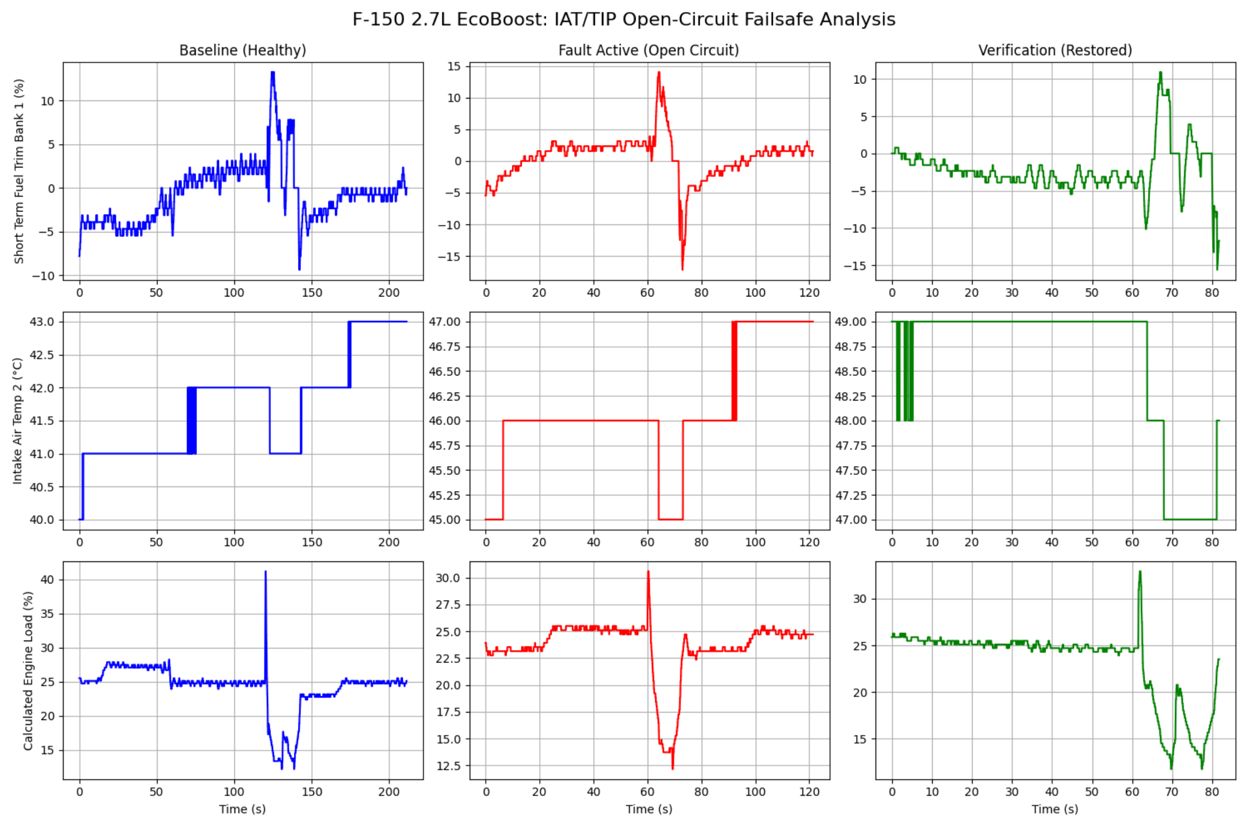

Figure 2.1: Side-by-side comparison of Baseline, Active Fault, and Verification telemetry.

SYSTEMS ENGINEERING & DIAGNOSTIC FINDINGS

- **Closed-Loop Fuel Trim Compensation:** With the IAT/TIP circuit open, the PCM lost its primary air-density input and could no longer trust its speed-density fueling calculation. Rather than risk a lean-detonation event under boost, it abandoned tight stoichiometric targeting and biased SHRTFT1 from a –0.68% baseline trim to +0.57% under active fault, commanding a safely richer mixture. The sharpest transient enrichment (exceeding +12% momentarily) lines up with the high-load excursion captured mid-run, showing the compensation strategy scales with instantaneous load rather than applying a single fixed offset.

- **Sensor Circuit Isolation:** Because IAT/TIP and IAT2 are physically separate sensors on distinct circuits, the secondary post-intercooler sensor (IAT2) remained fully active throughout the fault window, tracking real heat soak from 41°C at baseline to 48°C during the fault and verification runs. This redundancy was the key diagnostic clue: a live, trending IAT2 signal alongside a flat, defaulted IAT/TIP reading confirmed the failure was isolated to a single open circuit rather than a shared harness ground, a PCM-side fault, or a broader sensor-suite failure.

- **Verification Protocol:** Following physical restoration of the sensor connector, a Keep Alive Memory (KAM) reset was performed, and the DTC transitioned from Active to Stored/Historic. A subsequent run cycle confirmed IAT/TIP telemetry resumed active ambient tracking and SHRTFT1 stabilized to –2.2%, closely matching the pre-fault baseline. The MIL remained extinguished through the full verification cycle, confirming the repair fully restored closed-loop fueling control rather than merely clearing a stored code.

TECHNICAL SERVICE BULLETIN

TSB No: 26-1041	**Date:** July 2026
Group: Powertrain / Thermal Management	**Subject:** Charge Air Cooler (CAC) Thermal Saturation
Engine: 2.7L EcoBoost V6	**Affected Vehicles:** 2018 Ford F-150 **Status:** Technical Advisory

CONDITION

Some vehicles operating in high-ambient temperature regions (90°F+) may exhibit a perceived loss of top-end power, delayed throttle response, or sluggish acceleration during repeated Wide Open Throttle (WOT) applications or heavy towing. The Malfunction Indicator Lamp (MIL) will NOT be illuminated, and no Diagnostic Trouble Codes (DTCs) will be stored in the Powertrain Control Module (PCM).

TECHNICAL CAUSE

Engineering telemetry indicates the factory air-to-air Charge Air Cooler (CAC) can reach thermal saturation ("heat soak") under sustained maximum boost delivery (211 kPa / 16.2 psi). When post-intercooler Intake Air Temperatures (IAT2) exceed 45°C, the PCM actively retards ignition timing to prevent pre-ignition/detonation, resulting in an intentional reduction of torque output. This condition is often exacerbated by debris blocking the CAC fins or by mechanical binding of the Active Grille Shutter (AGS).

DIAGNOSTIC & SERVICE PROCEDURE

Do not replace the turbochargers, wastegate actuators, or high-pressure fuel pumps for this condition. Proceed with the following thermal diagnostic protocol:

Step 1: Visual and Mechanical Inspection

1. Inspect the front bumper lower intake aperture for physical obstructions (leaves, plastic bags, mud).
2. Using FORScan or equivalent diagnostic interface, command the Active Grille Shutters (AGS) from 0% to 100%. Verify smooth, unobstructed travel of the louvers.

Step 2: Dynamic Telemetry Validation

1. Connect a high-speed OBD interface and monitor the following PIDs: RPM, MAP, IAT2, WGATE_PRES, SPARKADV.
2. Bring the engine to normal operating temperature (ECT > 195°F).
3. Perform a 3rd-gear WOT sweep from 2,500 to 5,000 RPM in a safe environment.

4. Analyze the data log: If MAP reaches >200 kPa but IAT2 climbs rapidly past 45°C and SPARKADV drops, the CAC is experiencing thermal saturation.

Step 3: Corrective Action

1. If the AGS is binding, clear the obstruction or replace the shutter assembly.
2. Clean the exterior fins of the CAC using low-pressure water and compressed air (max 30 psi) to restore thermal transfer efficiency.

CORRECTIVE ACTION & VERIFICATION

Corrective action was completed per Step 3 of the diagnostic procedure above (obstruction clearance / AGS service and CAC fin cleaning). Verify the repair using the protocol below before returning the vehicle to service.

Perform a secondary WOT sweep. Ensure IAT2 recovery times have improved, and spark advance remains steady under peak boost pressure.